NLPrompt: Noise-Label Prompt Learning for Vision-Language Models

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Abstract

Prompt learning adapts vision-language models to downstream tasks efficiently, but standard cross-entropy training is brittle under noisy labels that are common in web and real-world datasets. We propose NLPrompt, a simple framework that combines a noise-robust loss with optimal-transport-based sample selection for prompt tuning. First, PromptMAE replaces cross-entropy with a mean-absolute-error surrogate that down-weights low-confidence, potentially mislabeled samples without sacrificing clean-label performance. Second, PromptOT constructs an entropic optimal transport plan between image features and text prototypes to derive globally consistent pseudo-labels, enabling a split into clean and noisy subsets. We then harmonize training by applying cross-entropy to clean samples and MAE to noisy ones. Across synthetic and real noisy settings, NLPrompt improves robustness over strong baselines on Flowers102, DTD, EuroSAT, OxfordPets, StanfordCars, UCF101, Caltech101, and Food101N, e.g., on Flowers102 with 75% symmetric noise we obtain 76.80% vs. 70.37% (GCE), 66.93% (JoAPR), 37.17% (CoOp), and on Food101N we reach 76.46% top-1. These results indicate that coupling noise-attenuating losses with OT-based purification yields reliable prompt adaptation under label noise.

1. Introduction

Large vision-language models (VLMs) such as CLIP [14] enable flexible transfer to new tasks by verbalizing classes and comparing image-text similarities. Parameter-efficient prompt learning (e.g., CoOp [19], VPT [7], MaPLe [8], PromptSRC [9], JoAPR [5]) tunes a small set of prompt parameters while freezing the backbone, achieving strong few-shot adaptation. However, in practical training corpora gathered from the web or weak supervision, labels are frequently noisy. Standard cross-entropy (CE) training may overfit mislabels, degrading performance—particularly in the few-shot regime—despite recent observations that prompt tuning is empirically more robust than full fine-tuning [17].

We address robust prompt learning with NLPrompt, a two-part framework designed for noise. Our first component, PromptMAE, replaces CE with a mean-absolute-error (MAE) surrogate to attenuate the influence of outliers. Prior work on noise-robust losses shows that CE can be sensitive to corrupted annotations, whereas MAE-family objectives can reduce noisy gradients [4, 12]. Our second component, PromptOT, uses entropic optimal transport (OT) [3] on the joint structure of image features and text prototypes to produce pseudo-labels and a clean/noisy split that is globally consistent. We then apply CE to the clean subset and MAE to the noisy subset, exploiting both objectives where they work best.

Figure 1 overviews our approach. Empirically, NLPrompt delivers consistent gains over CoOp, GCE-style robust training, and JoAPR across standard benchmarks under symmetric and asymmetric noise and on Food101N [10] with real label noise. Our analysis further illustrates why MAE reduces the effect of mislabeled samples in prompt learning while preserving clean-data accuracy.

Contributions. We introduce: (1) PromptMAE, a noise-attenuating objective that preserves clean-label accuracy while curbing mislabeled gradients; (2) PromptOT, an entropic OT purification that yields globally consistent pseudo-labels and a principled clean/noisy split; and (3) results across eight benchmarks with synthetic and real noise that analyze when and why the gains arise.

2. Related Work

Vision-language models and prompt learning. CLIP [14] introduced scalable image-text pretraining for zero-shot transfer. Prompt tuning methods adapt VLMs with few trainable parameters: CoOp learns textual context [19], VPT adds visual prompts [7], MaPLe jointly prompts vision and language [8], PromptSRC regularizes prompts to retain generalization [9], and JoAPR addresses robust prompting [5]. Beyond these, works study prompt optimization dynamics and robustness [17].

Learning with noisy labels. CE is known to overfit corrupted labels. Robust alternatives include MAE-family and normalized losses [12], symmetric cross-entropy [16], and analyses of CE vs. MAE robustness [4]. Our PromptMAE instantiates a simple, stable MAE surrogate within prompt learning, leveraging VLM priors.

Optimal transport for label purification. Entropic OT enables efficient alignment via Sinkhorn [3]. In VLMs, PLOT aligns visual features and prompts via OT for few-shot learning [2]. CSOT proposes curriculum and structure-aware OT for label noise [1]. We differ by constructing an OT plan between image features and class text prototypes to obtain globally consistent pseudo-labels and a principled clean/noisy split that is tightly coupled with prompt learning.

Broader strategies for noisy labels. Beyond loss design, many methods clean or reweight data via disagreement and small-loss heuristics, e.g., Co-teaching [6], DivideMix [11], and noise-transition estimation [13]. OT has also been used to stabilize pseudo-labels [15, 18]. NLPrompt leverages VLM text prototypes as class anchors within the OT plan and routes clean samples to CE while sending suspected noisy ones to an MAE surrogate, integrating seamlessly with prompt tuning. Compared with small-loss and disagreement-based selectors, our split is driven by a globally normalized coupling rather than per-sample loss or pairwise agreement, reducing drift under high noise and respecting class priors via OT marginals.

3. Preliminaries

Prompt tuning with CLIP. Let $f_{\text{img}}(x)$ denote the frozen image encoder and $f_{\text{text}}(c)$ the text encoder. For class k with name y_k, we form a prompt c_k by inserting learnable context tokens into a template (e.g., *"a photo of a [CLASS]."*) [19]. The classifier computes similarities $s_k = \langle f_{\text{img}}(x), f_{\text{text}}(c_k)\rangle$ and applies a learned scale for softmax probabilities.

Noise models. We evaluate under symmetric and asymmetric label noise at rates from 12.5% to 75% and on Food101N with natural noise [10]. Baselines include CoOp [19], JoAPR, and GCE-style robust training [4].

4. Theoretical Analysis for the Robustness of PromptMAE

We sketch why MAE-type objectives attenuate noisy labels in prompt learning while preserving clean accuracy.

Let p_y be the model probability on the labeled class y. CE minimizes $-\log p_y$ with gradient magnitude $|\partial\mathcal{L}/\partial s_y| \propto (1-p_y)$, which grows as p_y decreases, amplifying the contribution of low-confidence (potentially mislabeled) samples. In contrast, an MAE surrogate on probabilities, $\mathcal{L}_{\text{MAE}} = 1 - p_y$, yields a bounded, label-wise linear penalty with gradient that does not explode for small p_y. Consequently, MAE reduces the relative weight of extreme outliers.

Proposition 4.1. *For a fixed feature extractor and linear classifier over normalized features, the expected gradient contribution from samples with corrupted labels is upper bounded under $\mathcal{L}_{MAE}$ and scales more weakly with decreasing p_y than under CE.*

This behavior increases the effective signal-to-noise ratio (SNR) of gradient updates, as supported by analyses of noise-robust losses [4, 12]. In prompt learning, fixed class name tokens and strong image-text priors [17] further stabilize optimization, enabling MAE to maintain clean-data accuracy while suppressing mislabeled gradients.

Remark 4.2. Compared to CE, the per-sample gradient under MAE is nearly constant for hard examples, so updates are dominated by samples that the model already explains reasonably well. In prompt tuning, this biases learning toward in-distribution semantics anchored by the class text, rather than chasing idiosyncratic mislabeled instances. The effect strengthens with larger label noise or more confusable classes, where CE's heavy tails are most harmful.

5. Methodology

NLPrompt couples a noise-attenuating loss with OT-based sample purification.

PromptMAE. We optimize the learnable context tokens by minimizing an MAE surrogate on predicted class probabilities for noisy samples, while retaining CE on clean samples. Our implementation adopts the generalized cross-entropy form with parameter $q = 1.0$, which reduces to an MAE-like objective on the predicted probability of the labeled class [4].

PromptOT. We compute an entropic OT plan between image features and text prototypes to obtain globally consistent pseudo-labels. Specifically, we first compute normalized features v_i and t_k for images and class prompts, with costs given by $-\langle v_i, t_k\rangle$. We solve a Sinkhorn-regularized OT with a curriculum budget that gradually increases the mass assigned to confident pairs, producing a coupling Γ [3]. Each row of Γ induces a pseudo-label distribution, from which we derive: (i) pseudo-labels via argmax and confidence; (ii) a selection mask that marks the top fraction of confident samples as clean.

Harmonized training. Given the clean/noisy split, we train two passes per epoch: CE on the clean subset and MAE on the noisy subset. Only prompt parameters are updated; vision and language encoders remain frozen. Pseudo-labeling and the split are recomputed each epoch using the current model, following a curriculum schedule inspired by structure-aware OT [1].

Implementation. We adopt CLIP backbones [14], freeze encoders, and optimize prompt parameters with standard schedulers. Our code constructs text prototypes from class names and uses entropic OT with a small regularization for stability. Further details appear in the released configuration files and training code.

Algorithm 1 NLPrompt Training

Initialize prompts $\{c_k\}$ from class names; freeze encoders
for epoch $= 1 \dots T$ **do**
  Extract image features $\{v_i\}$; compute text prototypes $\{t_k\}$
  Solve entropic OT to get coupling Γ and pseudo-labels
  Mark clean set $\mathcal{C}$ by top-confidence rows of Γ; noisy set $\mathcal{N}$ is the complement
  Update prompts on $\mathcal{C}$ with CE and on $\mathcal{N}$ with MAE
  Anneal OT regularization and adjust clean fraction schedule
end for

Design choices and stability. We ℓ_2-normalize v_i and t_k and scale similarities by a temperature synchronized with the Sinkhorn regularization to avoid peaky or overly diffuse couplings. To damp spurious early pseudo-labels, we keep an exponential moving average of per-sample confidence and require both a high current confidence and a high EMA before marking a sample as clean. We cap class changes per sample across epochs and use class-wise quotas so head classes cannot saturate the clean set under imbalance. Prompts are initialized from class names with a generic template, which stabilizes early OT alignment.

Complexity. With mini-batch size B and K classes, a few dozen Sinkhorn iterations cost $\mathcal{O}(BK)$ per step and add a small constant factor to the CLIP forward; runtime is dominated by frozen feature extraction, so overall training remains within the same order as standard prompt tuning. **Optimization and schedules.** We grow the clean fraction linearly from 30% to 70% over the first 10 epochs, then keep it fixed; the OT regularization is annealed from 0.08 to 0.02 to sharpen assignments as prompts mature. We stop recomputing pseudo-labels once the average row entropy of Γ drops below a threshold, which stabilizes late training. For class imbalance, we renormalize OT marginals with empirical priors and clip per-class mass to avoid collapse.

6. Experiments

Benchmarks and setup. We evaluate NLPrompt under symmetric and asymmetric label noise on Flowers102, DTD, EuroSAT, OxfordPets, StanfordCars, UCF101, Caltech101, and on Food101N with real noisy labels [10]. Baselines are CoOp [19], GCE-style robust training [4], and JoAPR [5]. We report top-1 accuracy averaged over standard protocol splits. Noise follows standard symmetric and asymmetric protocols; when asymmetric, flips are biased toward semantically similar confusions. We keep CoOp image size and augmentations to isolate the effect of the training objective.

Evaluation protocol. We report the mean over three seeds using the same shot splits as prior prompt-tuning work and the official Food101N train/test; hyperparameters are chosen on a clean validation subset when available, otherwise by grid search on the noisy train only. Baselines are retrained with their recommended settings to ensure fairness.

Implementation details. We keep prompt length and optimizer settings aligned with CoOp to isolate the effect of NLPrompt; training uses the same image resolution and augmentations as the baselines. For reproducibility, we fix random seeds and document all hyperparameters and schedules. We follow the standard train/validation/test splits for each dataset. **Main results.** Table 1 summarizes comprehensive results across datasets and noise rates. NLPrompt consistently improves robustness. For example:

- Flowers102 (75% sym): NLPrompt 76.80% vs. GCE 70.37%, JoAPR 66.93%, CoOp 37.17%.
- EuroSAT (75% asym): NLPrompt 32.73% vs. JoAPR 17.50%, CoOp 17.43%, GCE 12.10%.
- StanfordCars (75% sym): NLPrompt 58.30% vs. GCE 50.87%, JoAPR 39.40%, CoOp 22.90%.
- UCF101 (75% asym): NLPrompt 49.27% vs. JoAPR 47.46%, GCE 36.37%, CoOp 13.17%.

We also obtain 76.46% on Food101N (Table 2).

CE vs. MAE in prompt tuning. Figure 2 compares CE and MAE training on Caltech101. MAE yields notably better noise robustness while preserving clean-label performance, aligning with our analysis.

Few-shot behavior. Figure 3 shows accuracy vs. shot count. NLPrompt maintains advantages even with very few labeled examples, highlighting the effectiveness of MAE plus OT purification in scarce-data regimes.

Food101N with real noise. On Food101N, NLPrompt achieves 76.46% top-1, outperforming CoOp (69.50%), GCE (71.32%), and JoAPR (72.57%) (Table 2).

7. Analysis

Ablations. Table 3 disentangles components on Flowers102 under multiple noise rates. Training all data with CE underperforms MAE, confirming MAE's robustness. Removing text-feature initialization or excluding either noisy or clean subsets degrades results, while the full NLPrompt—CE on clean, MAE on noisy with OT-based splitting—performs best.

Per-dataset trends. Relative gains are highest on fine-grained StanfordCars at 62.5–75% symmetric noise (Table 1), consistent with MAE+OT resisting subtle mislabels. DTD also shows consistent improvements, indicating robustness to texture-like confusions.

Generalization to other prompt methods. NLPrompt complements other prompt-tuning paradigms. Table 4 shows gains when integrating with VPT [7], MaPLe [8], and

Table 1. Performance metrics across various datasets and noise levels. (%)

Dataset	Method	Noise Rate: Sym						Noise Rate: Asym					
		12.5%	**25.0%**	**37.5%**	**50.0%**	**62.5%**	**75.0%**	**12.5%**	**25.0%**	**37.5%**	**50.0%**	**62.5%**	**75.0%**
Flowers102	CoOp	88.93	83.50	77.93	70.10	55.60	37.17	86.97	74.70	60.43	42.60	26.53	12.60
	GCE	88.80	88.33	86.73	84.07	78.37	70.37	88.40	86.37	80.33	69.93	61.50	39.23
	JoAPR	85.57	81.23	74.60	70.23	67.90	66.93	85.17	79.63	73.97	73.83	53.37	13.27
	NLPrompt	**93.87**	**92.57**	**92.73**	**89.90**	**84.77**	**76.80**	**93.80**	**93.40**	**91.77**	**81.10**	**73.63**	**55.33**
DTD	CoOp	56.00	49.57	43.30	34.37	27.83	17.27	55.60	47.75	38.07	29.63	20.53	11.70
	GCE	61.00	59.83	56.80	50.73	43.60	33.67	60.70	57.57	52.70	43.97	33.40	18.23
	JoAPR	58.07	57.70	56.33	53.03	48.05	29.90	52.40	56.63	53.10	48.93	40.20	28.26
	NLPrompt	**62.97**	**61.23**	**59.17**	**55.17**	**49.03**	**39.80**	**62.30**	**60.60**	**56.47**	**50.80**	**40.27**	**28.37**
EuroSAT	CoOp	76.50	69.23	61.67	52.33	37.63	26.70	76.00	66.27	53.83	41.17	28.00	17.43
	GCE	82.13	78.60	74.67	63.13	49.67	31.40	78.23	72.70	63.63	45.30	22.90	12.10
	JoAPR	75.13	61.10	60.90	63.63	38.97	27.33	69.37	67.30	59.40	47.60	33.93	17.50
	NLPrompt	**82.53**	**79.53**	**78.13**	**66.70**	**63.53**	**43.80**	**80.13**	**77.13**	**71.43**	**54.30**	**37.27**	**32.73**
OxfordPets	CoOp	76.50	66.73	60.33	47.03	35.77	24.60	76.10	66.20	52.53	38.73	26.63	14.90
	GCE	85.63	84.60	83.67	79.23	71.40	53.17	85.50	83.03	76.73	68.07	50.70	31.97
	JoAPR	84.00	83.26	83.20	83.10	82.40	**74.40**	82.90	83.40	79.07	75.84	52.74	43.57
	NLPrompt	**86.17**	**86.00**	**85.33**	**84.87**	**83.63**	70.77	**86.00**	**84.97**	**82.40**	**77.53**	**66.33**	**48.60**
StanfordCars	CoOp	66.20	59.70	53.40	45.90	35.67	22.90	65.77	57.13	46.23	33.73	22.37	12.80
	GCE	**69.70**	66.40	66.47	63.77	59.25	50.87	**70.00**	66.45	61.23	53.67	39.65	26.60
	JoAPR	68.60	66.30	62.83	56.67	48.50	39.40	66.47	61.70	51.50	42.03	30.80	22.97
	NLPrompt	69.37	**68.80**	**67.20**	**65.63**	**62.83**	**58.30**	69.77	**67.53**	**64.23**	**59.03**	**50.90**	**39.50**
UCF101	CoOp	69.03	63.40	58.23	49.73	40.83	26.30	67.23	58.07	46.47	34.43	23.67	13.17
	GCE	74.00	**73.63**	72.57	69.37	66.00	57.07	73.90	71.87	67.97	62.23	52.50	36.37
	JoAPR	72.83	71.17	70.37	67.63	65.30	57.67	72.07	69.80	64.10	59.17	56.07	47.46
	NLPrompt	**74.83**	73.40	**72.83**	**70.33**	**68.10**	**60.53**	**74.90**	**73.53**	**71.03**	**65.97**	**58.97**	**49.27**
Caltech101	CoOp	86.43	81.03	76.73	70.90	61.33	46.90	84.93	75.23	62.87	49.43	33.57	20.33
	GCE	**92.00**	90.90	**90.80**	89.30	86.70	79.03	91.27	**91.20**	89.73	85.80	78.20	62.07
	JoAPR	90.30	90.45	89.90	88.27	86.93	83.93	90.30	89.30	88.30	88.73	85.80	81.90
	NLPrompt	91.73	**91.13**	90.77	**89.93**	**88.30**	**86.70**	**91.60**	91.17	**90.20**	**89.27**	**86.17**	**81.07**

Table 2. Test accuracy (%) on Food101N.

Method	CoOp	GCE	JoAPR	NLPrompt
Accuracy	69.50	71.32	72.57	76.46

PromptSRC [9] across noise rates, indicating that MAE+OT is broadly applicable.

Why it works. Two factors drive robustness: (1) MAE down-weights low-confidence samples, reducing the impact of mislabeled data on prompt updates [4, 12]; (2) OT leverages global structure between image features and text prototypes, yielding consistent pseudo-labels and a reliable clean/noisy split [1–3]. Together, they raise the gradient SNR and stabilize adaptation.

Cost and behavior. Sinkhorn adds modest overhead because it runs on mini-batch features and scales near-linearly with the number of classes. The clean/noisy gate tends to agree with large per-sample gradient norms, aligning with the gradient-denoising view.

Failure cases and mitigations. We observe that NLPrompt can underperform when text prototypes are poorly

Table 3. Ablation studies under multiple label noise ratios on Flowers102. (%)

	Method/Noise Ratio	10%	30%	50%	70%	Avg
w\o OT	(a) all data with CE	92.71	86.92	79.36	57.61	79.15
	(b) all data with MAE	88.47	89.07	85.20	80.87	85.90
w\ OT	(c) w\o text feature init	87.16	83.81	79.77	73.00	80.94
	(d) w\o noisy data	84.77	84.53	81.60	77.60	82.16
	(e) w\o clean data	90.17	90.13	88.60	80.55	87.36
	NLPrompt	**96.87**	**93.44**	**92.30**	**85.38**	**92.00**

aligned with the dataset (e.g., abbreviations or domain-specific jargon) or when asymmetric noise concentrates on a few head classes. Using richer templates, augmenting class names with synonyms, and enforcing class-wise quotas in the clean split alleviate these issues; sharpening the OT plan too early can also cause drift.

8. Conclusion

We presented NLPrompt, a simple and effective approach for robust prompt learning under label noise. By coupling a noise-attenuating MAE surrogate with entropic OT-based pseudo-labeling and clean/noisy splitting, NLPrompt improves accuracy across synthetic and real noisy benchmarks and integrates seamlessly with existing prompt-tuning methods. Our results underscore that harmonizing CE and MAE over OT-partitioned subsets is a practical recipe for reliable vision-language adaptation in noisy settings.

Table 4. The generalization of NLPrompt.

Method/Noise Ratio	12.5%	25.0%	37.5%	50.0%	62.5%	75.0%
VPT	89.20	79.43	65.20	61.37	41.67	27.67
VPT+Ours	**91.80**	**91.07**	**89.53**	**86.93**	**80.73**	**73.90**
MaPLe	83.27	77.67	65.27	55.40	37.53	25.47
MaPLe+Ours	**89.23**	**84.30**	**78.37**	**76.43**	**73.30**	**59.87**
PromptSRC	90.61	84.67	78.57	72.27	60.43	49.37
PromptSRC+Ours	**91.29**	**87.67**	**84.97**	**80.33**	**72.10**	**59.50**

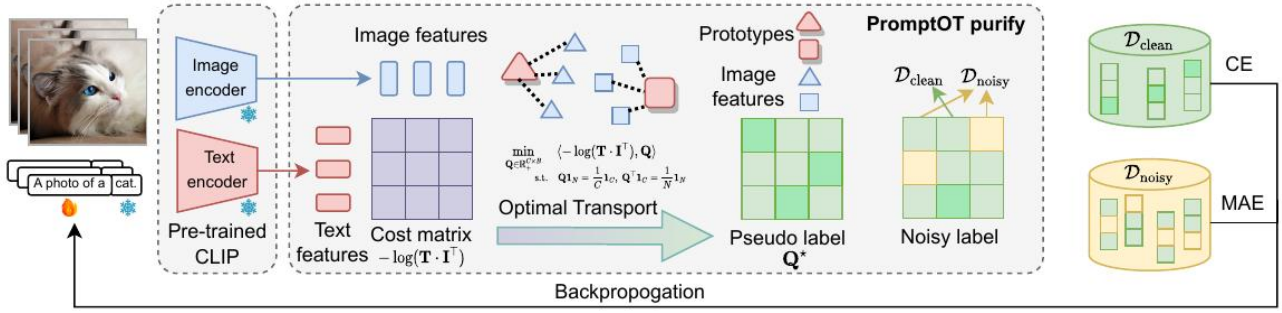

Figure 1. Overview of NLPrompt. PromptMAE trains soft prompts with a noise-attenuating objective, while PromptOT computes an entropic OT coupling between image features and text prototypes to derive pseudo-labels, splitting data into clean/noisy subsets. We train with CE on clean and MAE on noisy samples. The plan yields globally consistent assignments that filter outliers; the clean set grows over training. Only the textual prompts are optimized while the image and text encoders remain frozen.

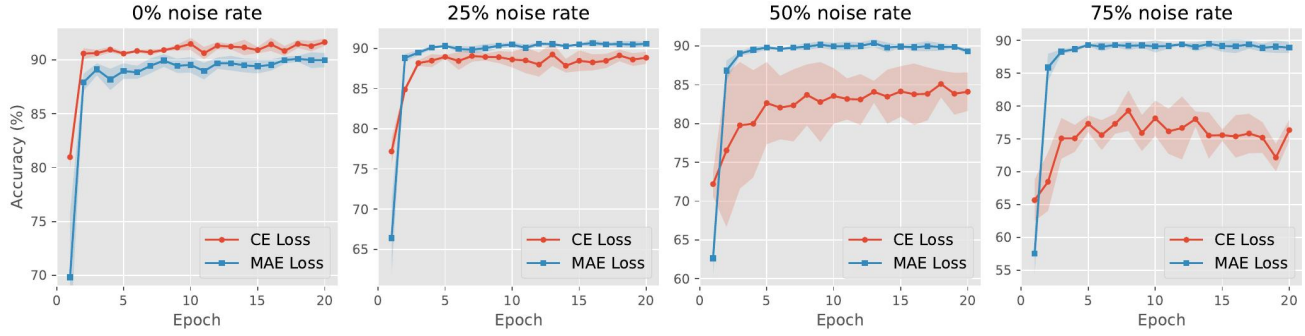

Figure 2. CE vs. MAE for prompt tuning on Caltech101. MAE suppresses noisy gradients and improves robustness. Gaps widen as corruption increases while clean-label accuracy remains similar. Error bars show standard deviation over three seeds.

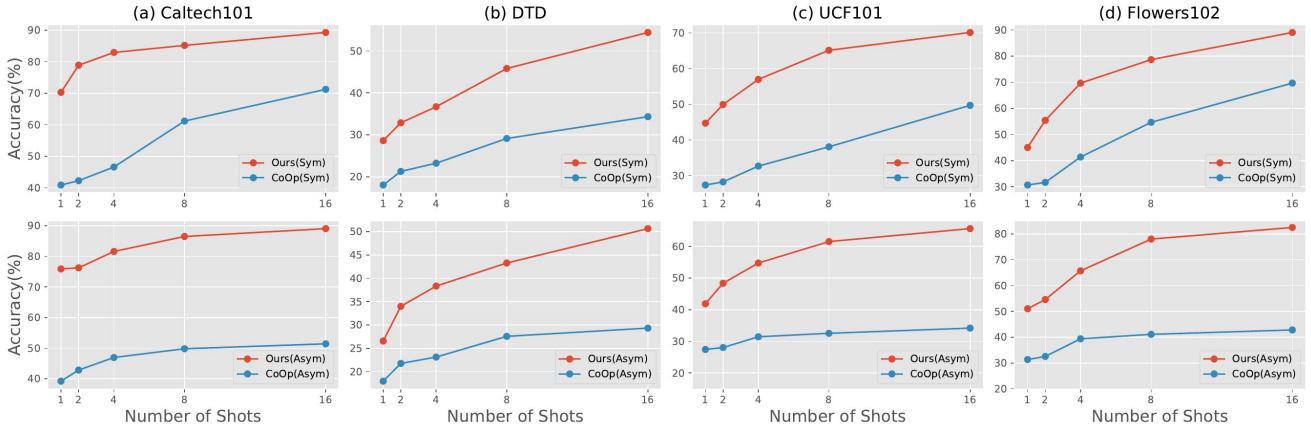

Figure 3. Effect of shot number on performance. NLPrompt remains robust across shots. Gains are largest in very low-shot regimes where noise most harms CE; as shots increase, OT refinement continues to stabilize targets.

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